

WikiContrib Product Requirements Document

Version 1.0 - core analytics complete - final delivery stage

Document field	Value
Project	WikiContrib - Wikipedia Contributor Impact
Document	Product Requirements Document (PRD)
Version	v1.0
Owner	Han Zhu (Han-ZhuAI)
Date	1 July 2026
Code baseline	main - merge commit bc08e4b
Status	Formal design baseline

Revision summary

- Defines the product scope, decision rules, termination conditions, exceptions and acceptance criteria against the current main branch.
- Unifies token-level diff, provenance chaining, wikitext element classification, Talk-page PageRank and four-axis scoring into one auditable workflow.
- Separates implemented analytical capabilities from pending report, video, README and v1.0 release work.
- Records the verified quality baseline: 245 tests, 95.45% total coverage, and passing Python 3.10 and 3.12 GitHub Actions jobs.

1. Product background and definition

Wikipedia edit counts measure activity but do not reliably measure impact. A high-value contribution may be durable prose, structural maintenance, a revert of damaging content, or a Talk-page intervention that shapes consensus. Byte deltas and current-page attribution also lose temporal evidence.

WikiContrib is an explainable contributor-impact analysis tool for research and academic evaluation. Given a Wikipedia article title, it reads public article and Talk revision histories, creates auditable evidence, builds four-dimensional contributor profiles, and exports deterministic rankings, JSON and charts.

PRODUCT PRINCIPLE Every ranking must be traceable to source revisions, raw metrics, normalized dimensions, explicit weights and individual weighted terms. The composite score is a policy, not an objective measure of human worth.

2. Target users

- Course evaluator - verifies genuine implementation, tests, CI, real-article evaluation and reproducible outputs.
- Researcher - compares contributors under explicit impact policies and tests whether conclusions are sensitive to weight choices.
- Wikipedia analyst - distinguishes content creation, maintenance, surviving authorship and discussion influence.
- Project maintainer - needs clear module boundaries, cache behavior, quality gates and release criteria.

3. Use cases

Use case 1 - evaluate the main contributors to one article

An evaluator analyzes 'Alan Turing' with a 50-revision limit. The system labels the result as an earliest-50 historical slice, ranks contributors, and exposes volume, additive, persistence and discussion dimensions with their weighted terms.

Use case 2 - test alternative scoring policies

A researcher emphasizes volume, additive work, persistence and discussion in separate runs. The system reports winner changes, top-k overlap, Spearman rho and rank shifts so that ranking sensitivity is visible.

Use case 3 - audit a high score

An analyst moves from the composite result back to raw metrics, revision diff, token provenance, semantic element deltas and Talk reply edges, verifying that the result is not merely edit count or automated-account activity.

4. Core workflow

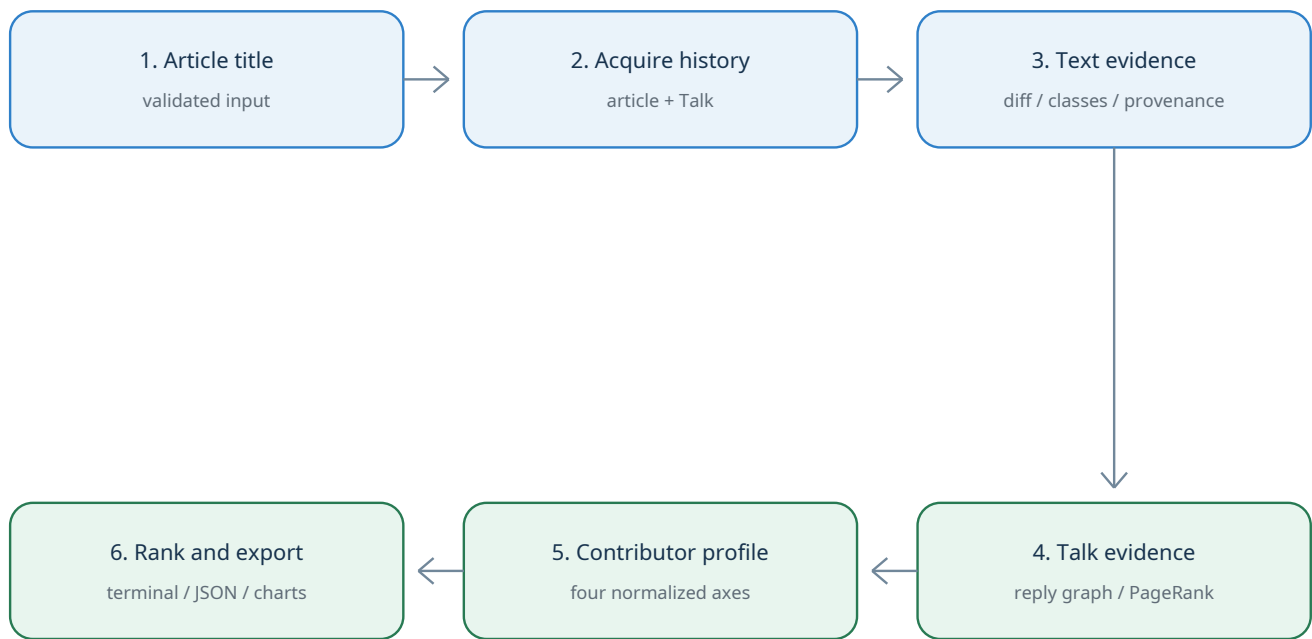


Figure 4-1. End-to-end WikiContrib workflow

The analytical path is deterministic. Text attribution, edit classification, PageRank, normalization and scoring are rule-based; identical inputs and scope must produce identical results.

5. Functional requirements

ID	Capability	Testable requirement	Status
FR-01	Revision acquisition	Fetch article and Talk revisions through MediaWiki Action API, follow pagination and return oldest-first records.	Complete
FR-02	Scope and cache	Default to earliest 500 revisions, support --all-revisions, cache source responses and label every capped run as a historical slice.	Complete
FR-03	Token diff	Calculate added, removed, net, gross and churn between consecutive wikitext states; separate Unicode words from symbols.	Complete
FR-04	Edit classification	Classify additive versus maintenance edits using diff shape, summary keywords, minor flag and identity-revert evidence.	Complete
FR-05	Provenance	Track original token authorship through the revision chain, including exact reverts and conservative partial restoration.	Complete
FR-06	Wikitext semantics	Assign every changed word to at most one of prose, heading, reference, template, table, category or link.	Complete
FR-07	Talk analysis	Parse threads, signatures and indentation; create replier-to-replied-to weighted edges and deterministic PageRank with damping 0.85.	Complete
FR-08	Contributor profile	Join article-only, Talk-only and cross-surface users; retain raw metrics and calculate four normalized axes.	Complete
FR-09	Composite score	Normalize non-negative weights, expose every axis-times-weight term, rank deterministically and explain the result.	Complete
FR-10	Automation handling	Record candidate confidence, reasons and evidence revisions; exclusion affects ranking only and never deletes source evidence.	Complete
FR-11	Outputs	Provide terminal tables, schema-versioned JSON, atomic writes, full JSON with top-N display, and publication-ready PNG charts.	Complete
FR-12	Evaluation	Compare balanced and four one-axis-emphasis policies using winner, overlap, Spearman rho and rank-shift metrics.	Complete
FR-13	Final delivery	Publish the English report, 3-5 minute demonstration, final README status and v1.0 release.	Pending

6. Analysis-scope decision rules

These rules play the same role as intent-decision criteria in the Data Agent PRD: they determine how one run is interpreted and must be mechanically testable.

Condition	Decision	Required behavior
No revision scope supplied	Bounded default	Analyze the earliest 500 revisions and label every output <code>historical_slice=true</code> ; do not claim current-page coverage.
--all-revisions supplied	Full-history mode	Traverse all accessible revisions and warn that runtime and API load may be materially higher.
Talk page missing or no valid signatures	Discussion evidence unavailable	Continue article analysis, set the discussion signal to zero and emit a scope warning.
Username has automation indicators	Automation candidate	Retain all diff and provenance evidence; apply confidence-based eligibility only at ranking time.

Condition	Decision	Required behavior
Negative weights or zero total	Invalid scoring policy	Reject the run with a specific validation error; do not silently substitute default weights.
User identity hidden	Identity unavailable	Use an anonymous or hidden placeholder and do not infer a real person.

7. Metric and scoring definitions

DEFAULT FORMULA $\text{Impact}(u) = 0.25 \times \text{Volume}(u) + 0.25 \times \text{Additive}(u) + 0.25 \times \text{Persistence}(u) + 0.25 \times \text{Discussion}(u)$

Axis	Normalization	Interpretation
Volume	$\log(1 + \text{gross_words_u}) / \log(1 + \text{max_gross_words})$	gross_words = words_added + words_removed; captures total touched text, including rewrites.
Additive	$\text{additive_edits_u} / \text{article_edits_u}$	Equivalent to $1 - \text{maintenance_ratio}$; the share of article edits classified as additive.
Persistence	$\log(1 + \text{surviving_words_u}) / \log(1 + \text{max_surviving_words})$	Word tokens attributed to the contributor that remain in the final revision of the selected scope.
Discussion	$\text{pagerank_u} / \text{max_pagerank}$	Relative weighted Talk reply-graph PageRank; edge direction is replier to replied-to user.

- All axes are constrained to 0..1. User-supplied non-negative weights are normalized to sum to one.
- Results sort by descending score, with username order as the deterministic tie-break.
- Wikitext element classes are explanatory in v0.1.0 and do not directly alter the four-axis score.
- Scores are comparable only within one article and selected revision scope.

8. Goals by phase

Current MVP goal

- Complete the article-plus-Talk evidence chain and produce an explainable contributor ranking.
- Maintain automated tests and at least 95% total coverage on the main branch.
- Provide reproducible evaluation across at least three materially different Wikipedia articles.

Final v1.0 delivery goal

- Publish an English report based on fixed evaluation inputs, generated charts, limitations and sensitivity findings.
- Record a 3-5 minute demo showing one default analysis and one custom-weight comparison.
- Update README status and publish the final tag and GitHub release.

9. Acceptance criteria

ID	Criterion	Current evidence
AC-01	A valid title runs end to end from revision acquisition to ranking, JSON and charts.	Pass
AC-02	Exact reverts restore original provenance; ambiguous short re-additions remain conservative.	Pass
AC-03	Each changed word receives at most one wikitext element label.	Pass
AC-04	Talk signatures and indentation produce reply edges and normalized PageRank.	Pass
AC-05	Every ranked result exposes all four axes and weighted terms.	Pass
AC-06	Alternative weights produce reproducible stability statistics on real articles.	Pass
AC-07	The main test suite meets the 95% coverage gate.	245 tests; 95.45%
AC-08	Python 3.10 and 3.12 GitHub Actions jobs complete successfully.	Pass
AC-09	Committed evaluation covers Alan Turing, Kimchi and Python.	Pass
AC-10	Final report, demo video, README status and v1.0 release are published.	Pending

10. Termination conditions and exception handling

10.1 Termination conditions

- Revision acquisition ends when MediaWiki has no continuation token or the explicit revision limit is reached; the reason is stored in scope metadata.
- Provenance ends after all selected revisions are processed and surviving ownership is summarized at the final selected state.
- Talk PageRank ends at convergence or the implementation iteration bound and must remain deterministic.
- Scoring ends after every eligible profile has four normalized axes and weighted terms; automation eligibility is applied afterwards.

10.2 Exception list

Exception	Required behavior
Article not found / invalid title	Stop with a clear error and do not create a plausible-looking result artifact.
MediaWiki timeout or HTTP error	Report the failure and preserve valid cache state; do not silently return an incomplete result.
Talk page absent	Continue article analysis, set Discussion to zero and write a warning.
History truncated	Keep a historical-slice label in terminal, JSON and chart titles.
Output path not writable	Fail the atomic write and leave no half-written file.
All candidates excluded	Retain audit evidence and state why the ranking is empty.

11. Command-line interaction and output contract

The current product interface is a CLI. Interaction quality is defined by explicit scope messages, structured tables, reproducible commands and stable output schemas rather than a web UI.

Output	Required contents	Primary consumer
Terminal	Scope warning, metric leaderboards, semantic deltas, automation review and composite breakdown.	Evaluator
JSON	Schema version, metadata, raw metrics, feature vector, weights, weighted terms, rank and explanation.	Research / audit
PNG charts	Composite stack, role balance, activity timeline and radar profiles.	Report / presentation
Markdown evaluation	Cross-policy winners, top-k overlap, Spearman rho and rank shifts.	Final report

12. Non-functional requirements

- Reproducibility - deterministic sorting and tie-breaking, cached inputs and versioned outputs.
- Transparency - raw signals, normalized dimensions, weighted terms, scope limits and automation reasons remain visible.
- Reliability - MediaWiki pagination, timeout handling, explicit errors and atomic output writes.
- Performance - bounded default history and explicit full-history mode.
- API etiquette - descriptive User-Agent and controlled request behavior.
- Quality gate - Ruff, compile checks, pytest, coverage at or above 95%, and GitHub Actions.
- Privacy - public revision data only; hidden identities remain hidden.

13. Current status and release boundary

COMPLETE Token diff, edit classification, provenance, wikitext elements, Talk reply graph/PageRank, four-axis scoring, sensitivity evaluation, JSON/charts, automation candidates and CI.

PENDING Final English report, 3-5 minute demo video, final README status, delivery pull request and v1.0 release.

14. Out of scope

- Claiming that the composite score is an objective or universal measure of editor quality.
- Replacing Wikipedia governance, administrator judgement or bot policies.
- Inferring that a Talk post causally produced a later article edit.
- Building a relational database or multi-user web platform in v0.1.0.
- Direct absolute score comparison across different articles.